

MATH 4/51103

Intro to Abstract Algebra, II

Spring 2026

Professor John Bergdall
bergdall@uark.edu

MWF 8:35 - 9:25 AM
Classroom: HOEC 217

Overview

This is a second course in abstract algebra. We will develop the fundamental algebraic laws that govern polynomial equations. We do this by creating connections between algebraic numbers and symmetry groups in algebra.

Learning goals

Study the algebra and structure of polynomials.

Create techniques to build algebraic fields.

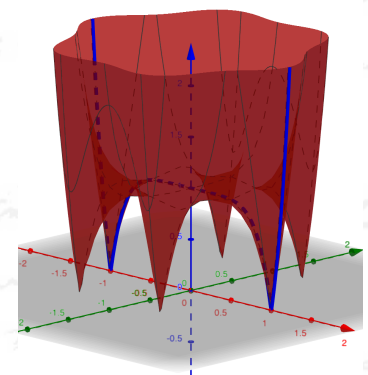
Exploit symmetry to universally solve equations.

Analyze and practice mathematical discourse.

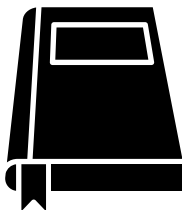
Try it out!

The roots of unity are complex solutions to the equation $z^n - 1 = 0$. You can visualize them by plotting the values of $z^n - 1$ using the Geogebra app <https://www.geogebra.org/m/qvqpd9yz> and looking for the stalactites that tip onto the complex plan.

(Pictured to the right: the plot for $n = 6$.)



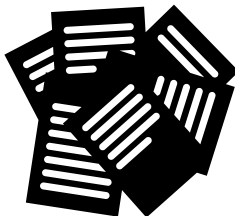
Basic resources



Textbook

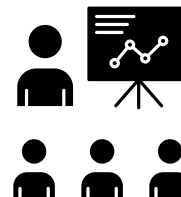
Abstract Algebra (4th ed.)
by Beachy and Blair
(Rent prints/digital: \$46/49*)

*If costs limit your course access, email me.



Course materials

Materials posted to Blackboard.
Assignments on Gradescope.
(Free)



Office hours (SCEN 353*)

Mon 1:00 - 2:00
Wed 1:00 - 3:00
Appointments and/or Zoom

*I have office chocolate for students.

Tasks and grades

Course tasks are divided into *Supporting Tasks* and *Performance Tasks*. Supporting Tasks represent class work and foundational reading. Performance Tasks represent chances to show your understanding of concepts and techniques. **Further details on pages 3-4.**



Supporting Tasks

Class Participation and Reading
Checks



Performance Tasks

Exercise Sets and Mastery
Assessments

Turn-in

You will have two frequent kinds of assignments to turn-in.

- **Reading Checks** are due weekly on **Sunday evenings at 11:59 PM.**
- **Exercise Sets** are due weekly on **Wednesday evening at 11:59 PM.**

Revisions

Revisions reinforce learning. Except for the final, all tasks may be revised. (Further details given in-semester.)

- **Reading Checks revisions** are due weekly on **Wednesday at 11:59 PM.**
- **Exercise Sets revisions** are due **36 hours** after the initial grades are returned.
- **Exam revisions** will take place in the **week after initial grades are returned.**

Late work

Due dates help keep your learning on track. But things come up. *You may turn work in late as part of the revision process.* However, you lose your chance to revise the work.

Supporting Tasks



In-class work gives you a voice in your learning. **Reading Checks** measure understanding of readings that frame class discussions.

In-class work (equal weight per task)

- **Be there.** Regularly attend meetings and work on tasks with peers.
- **Explain.** Regularly present your findings from the tasks, as a small group, at the board. You will do this once per week on average.

Details. In-class presentation of mathematics will be an *everyday* activity in this course. Presenting can be stressful, but students will have access to in-class work time, office hours, and appointments. For comfort, I *strongly recommend* trying your arguments on me before presenting!

- *End of class.* For the final portion of each meeting, students will work together in small groups on tasks related to that day's discussion.
- *Start of class.* For the initial portion of each meeting, groups will present their findings on the prior meeting's assigned tasks. The *group* will be graded based on the rubric found on p. 5.

Revision specifications. You may revise in-class presentations by providing written corrections. Group revisions is preferred. Individual revision is possible in case a consensus cannot be reached. If you miss your group's presentation, you may make it up with "leftover" tasks.

Reading Checks (equal weight per question)

- **Two questions** per reading, compiled into a weekly check.
- **Responses** in timed (10 min per Q) assignment on Gradescope.
- **Revisions** will include complete reasoning, in writing.



Details. Reading Check questions are *multiple choice* or *select all*.

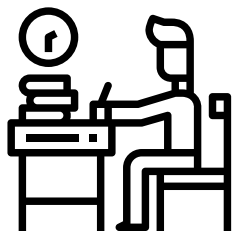
- Before the Check, you will know a prompt for the question and you will have been to the lectures for that week. You will not know the precise questions.
- Checks are open book/notes. You *may not* use generative AI. **See page 15 for AI guidelines.**
- The Checks are due **Sunday evenings at 11:59 PM**. Outcomes will be returned Monday morning.

Revision specifications. Here is how you will "revise your Reading Checks. **See page 12-13 for examples.**

- *Multiple choice.* Identify the correct answer along with a detailed explanation for why it is correct.
- *Select all.* First, identify the correct and incorrect options. Second, explain in detail why the correct ones are correct *and* why the incorrect ones are incorrect. You must do both steps!
- Complete and accurate revisions **earn back full points**. Some explanations may earn a half point.
- Revisions for a Sunday Reading Check are due the following **Wednesday at 11:59 PM**.

Performance Tasks

Exercise Sets let you expand on concepts and techniques practiced in class. **Mastery Assessments** are timed, in-class, long answer assignments that measure proficiency and mathematical discourse.



Exercise Sets (equal weight per question)

- **1-2 questions** per class lesson due weekly.
- **Turned in** electronically through Gradescope.
- **Collaborate!** As long as you attribute credit to your source(s).

Mastery Assessments

- **In-class essays** focused on specific learning goals and skills.
- **Two essays** per hour. Possible prompts available ahead of time.
- **Midterm dates:** Friday 2/20 (Week 6), Friday 4/10 (Week 13).
- **Final exam:** TBD in HOEC 217.



Details. Exercise Sets and Mastery Assessments are carefully designed based on learning goals.

- Problems on Exercise Sets will expand on some part of the in-class discussion and Supporting Tasks. They will often require further analysis and/or applying the analysis in a novel way.
- Exercise Sets are due **Wednesdays at 11:59 PM**.
- On Exercise Sets, you *may not* use generative AI to answer questions. **See page 15 for AI guidelines.**
- For midterms, I will post prompts at least one week prior to the test date. On the day of, you will have three prompts and you will respond to two of them. For the final, you will have to answer three prompts based on five options.

Revision specifications. First, when revising Performance Tasks, your new scores replace your old score!

- **Exercise Sets.** To revise a question on a Exercise Set, you should turn in a new solution along with an indication of your prior score.*
 - We will assess the work as if we never saw the prior answer. **See page 14 for examples.**
 - The revisions are due at 11:59 PM no earlier than 36 hours after the initial grades are returned, but I do not make revisions due on Friday or Saturday nights. Deadlines announced in class.
- **Midterms.** Revising your midterm essays is an in-person experience. I will re-award points based on your ability to fill in gaps in your original essay explain yourself.
 - **Step #1.** Prepare (for each essay) to explain to me a corrected version of the essay you wrote.
 - **Step #2.** Meet with me to present your explanation and answer spontaneous questions I have about your presentation and/or what you originally wrote.

* Your prior score is needed to simplify the logistics of revisions. Please, please, include your prior score so I do not have to look it up.

Grading Rubric

Your Performance Tasks and in-class Supporting Tasks will be graded using a 6-point rubric. The rubric will be strictly applied. Remember, revisions are always possible! **Examples of this rubric being applied are on pages 10-11.**

Details. Excelling in upper level courses means understanding concepts and mechanics. It also requires presenting answers in easy to follow fashion, with clear explanations. Our rubric emphasizes all of this.

	0 points	1 point	2 points
Concepts or strategy (up to 2 points)	The answer misidentifies or misuses crucial concept(s) or applies unproductive strategies.	The answer identifies and uses a crucial concept or attempts a productive strategy.	The answer identifies or applies <i>all</i> crucial concepts and succeeds in carrying out a productive strategy.
Mechanics or computation (up to 2 points)	The answer has significant or recurring issues with its mechanics or computation.	The answer shows proficiency in some mechanics and computation.	The mechanics and computations in the answer are flawless (and of the intended difficulty level*).
Grade-ability (up to 1 point)	The solution is not for the right problem, its writing is illegible, or grading is impossible for another reason.	The solution is for the right problem and the writing is neat enough to assess the work.	* "...intended difficulty level" means you don't simplify the problem to do an accurate calculation. See Solution #4 in the example on page 11.
Presentation (up to 1 point)	The solution omits vital writing necessary to explain the logic and reasoning needed for mathematical discourse.	The solution is written in mathematical discourse, with complete sentences giving all necessary explanations.	

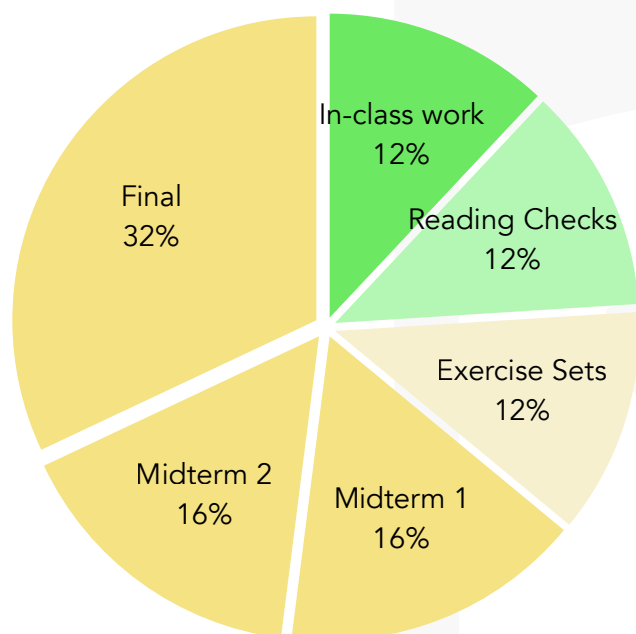
Originality. Your written work ***must be in your own words and reflect your own understanding***. If a student turns in an answer that uses approaches, results, terminology, etc. that are not consistent with the context of the course, they will receive a "temporary 0" on the problem. The 0 can be removed upon meeting with me to discuss the origin and logic of the solution. **See page 15 for AI guidelines.**

Essays. Your midterm and final essays be scored out of 10 rather than 6, with me doubling up on both concepts/strategy and mechanics/computation points.

Final grade

Your course grade will be derived from your Task scores according to the percentages in the pie chart below. The weighted scores determine your final grade according to the table on the right. *I reserve the right to make the gradelines more lenient.*

Task-by-Task weights for final grade



Gradelines

Grade	Minimum weighted score
A	90%
B	80%
C	70%
D	60%

Rationale. The 12% measurements are based 1% per week in each category, after omitting two midterm weeks, spring break, and the final week.

Rubric interpretation. The Performance Tasks are graded on a 0-6 scale and yet the gradelines are on a 90/80/70/60 scale. Here is one way to think about the Performance Tasks scores.*

- **A-level Performance Tasks = average 5.4.** So, half the time your work is excellent (earning 6's) and the other half there is at least excellence in concepts or mechanics (earning 5's).
- **B-level Performance Tasks = average 4.8.** So, nearly all the time your work is at least excellent in one of mechanics or concepts but maybe not the other (earning 5's).
- **C-level Performance Tasks = average 4.2.** Your work routinely shows proficiency in mechanics and concepts (earning 4's).

* This assumes you do the right problems and present your explanations in a clear fashion.

Survey bonuses

The final exam is initially weighted 2:1 against a midterm, but the ratio should be closer to 1.5:1. So, during the term I give *three* surveys: a pre-semester survey, a midterm survey, and the final course evaluations. Each survey in which the class reaches 80% participation is worth an automatic 2% coming out of the final exam. If all three meet that goal, an extra 2% will be counted.

Tips for success

I designed the course for you to pass. The revision process puts you in control of your grade. Everyone will need some help at some point in the semester. So, here are a few tips and resources you can pursue.

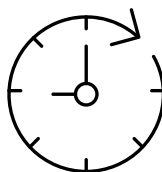


Message!

Write for help.



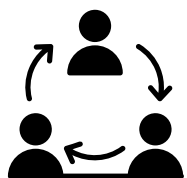
Ask! Me early and often.



Budget! Take your time.



Read! The text is a guide.



Collaborate!

With peers.

More tips (roughly in an order I follow)

- Take a break. Walk away.
- Explain to someone why you are stuck.
- Work out a simpler example.
- Search the internet.
- Ask an AI assistant. **See page 15 for AI guidelines.**
- Keep going!
- Tutoring is available in the Student Success Center in CORD.

Further Resources

Appointments

Office hours will not always fit your schedule. Email me to set appointments. In your initial message, please explain (i) what you want to talk about, (ii) *your* availability and (iii) if meeting in person is important (Zoom is possible).



Email

Your questions are important. Email me or post to Blackboard. (I may ask you to do this anyways.) Responses may take 24 hours, but send a reminder if I do not reply.

Access

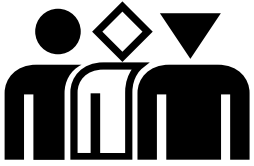
I want you to pass this class, with success. If you find resources not being provided, tell me and I will work with you to fix the issue. For some students, the Center for Educational Access (cea.uark.edu) coordinates student accommodation requests. Please request your accommodation letter early semester and meet with me to discuss it.



Class formation

Names and pronouns

You deserve to be called how you want. Let me know your preferred name and pronouns any time. When meeting one another, reintroduce yourselves if necessary. There is no reason to not know how someone is called.



Diversity

I pledge attention to your identity and experience, regardless of age, background, beliefs, ethnicity, gender identity and expression, national origin, racial identity, religious beliefs, sexual orientation, and any other visible or non-visible categories. You are expected to do the same for me and your peers.

Academic integrity

I trust my students to follow (i) the University of Arkansas Academic Integrity Policy and (ii) my own explicit assignment instructions. When you have questions about a boundary or rule, ask me rather than assuming you know the answer.



Technical policies

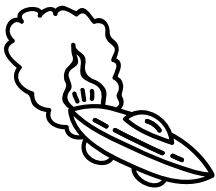
Inclement weather

If the University closes due to inclement weather, our course will meet asynchronously in the form of recorded lecture posted to Blackboard. Assessments, presentation, or due date will be postponed until the next day the University is open.



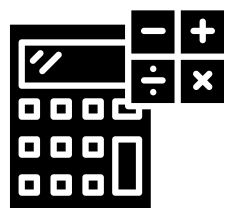
Class cancellation (emergencies)

If class is cancelled on short notice, I will work with the math office to try to find a substitute. If a substitute is found, class will be held. If not, alternative learning materials and assessments will be supplied. Regardless, an announcement will be sent promptly via email and posted to Blackboard.



Recordings and devices

I record class meetings using in-room recording systems and notes are live streamed via the OneNot (part of Office 365). If you use your devices (phone, tablet, etc.) to participate in class, you are expected to use those devices productively. Please tell me if someone's device is distracting you from your learning.



Calculators

A simple non-programmable calculator is not required. However, they are allowed and may give you confidence in your calculations during exams.

Schedule

	Wk.	Dates	Topic(s)	Reading
Study the algebra and structure of polynomials.	1	Jan 12-16	Foundations of fields and polynomial, review of complex numbers	Sec 4.1, A.5
	2	Jan 19-23	Roots of unity, polynomial factorization	Sec A.5, 4.2
	3	Jan 26-30	Unique factorization, isomorphisms, congruences	Sec 4.2-4.3
Create techniques to build algebraic fields.	4	Feb 2-6	Factor fields, tests for irreducibility	Sec 4.3-4.4
	5	Feb 9-13	Introduction to rings and homomorphisms	Sec 5.1-5.2
	6	Feb 16-20	Midterm break: review, problems, essays	—
Exploit symmetry to universally solve equations.	7	Feb 23-27	Algebraic numbers, degrees	Sec 6.1-6.2
	8	Mar 2-6	Tower law, explicit examples	Sec 6.2
	9	Mar 9-13	Splitting fields and the extension lemma	Sec 6.4
Analyze and practice mathematical discourse.	10	Mar 16-20	Uniqueness of splitting fields, separability, finite fields	Sec 6.4-6.5 Sec 8.1
	11	Mar 23-27	Spring break	—
	12	Mar 30-Apr 3	Galois groups, primitive elementary theorem	Sec 8.1-8.2
	13	Apr 6-10	Midterm break: review, problems, essays	—
	14	Apr 13-17	Fundamental theorem of Galois theory	Sec 8.3
	15	Apr 20-24	Computing Galois groups, radical extensions	Sec 8.3-8.4
	16	Apr 27-29	Solvability	Sec 8.4

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Performance Task Rubric

Here we provide examples for how Performance Tasks are graded. We chose problems from calculus, hoping that you will get the general idea without being confused about the mathematics.

Question.

Determine the area between the curve $y = \sin(x)$ and $y = \cos(x)$ over the x -range of $[0, \pi/4]$.

As a grader, I anticipate a student will convert the question into an integral calculation by subtracting one function from the other. Then, they will carry out the integral calculation by finding antiderivatives.

Solution #1.

$$\cos(x) \geq \sin(x) \text{ on } [0, \pi/4]$$

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = \sin(x) + \cos(x) \Big|_0^{\pi/4} = \sqrt{2} - 1 \approx 0.414...$$

Solution #2. We find the area by calculating an integral. Since $y = \cos(x)$ lies above $y = \sin(x)$ on $[0, \pi/4]$ (I graphed it), the relevant integral is:

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx.$$

Now we calculate:

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = \sin(x) - \cos(x) \Big|_0^{\pi/4} = 0 - 1 = -1.$$

Since areas are positive, the area between the curves is 1.

Grading solution #1: 4/6.

- *Concept.* 2/2. The solution finds an integral, after identifying which function to subtract from the other.
- *Mechanics.* 1/2. The integral is calculated completely accurately but the inequality is not explained.
- *Grade-ability:* 1/1. The writing (typing) is neat.
- *Presentation:* 0/1. There is no writing that indicates the goal, outcome, or approach of the solution. (This include the mysterious " $\cos(x) \geq \sin(x)$ " just floating on the page.)

Important point: the question asks for an *area* between two curves. The solution just calculates an integral. The missing explanation is how to connect the two.

Grading solution #2: 4/6.

- *Concept.* 1/2. The solution finds an integral to calculate. However, if we believe the setup is right then there is a conceptual error in the calculation turning up the answer -1. That is inconsistent conceptual knowledge.
- *Mechanics.* 1/2. The integral is not calculated accurately because there is a mistake in the antiderivative of $\sin(x)$. (Explaining you graphed the functions is fine, in this case, so long as you are right.)
- *Grade-ability:* 1/1. The writing (typing) is neat.
- *Presentation:* 1/1. The writing narrates in a clear way how the question is being answered.

This may take getting used to, but I consider conceptual gaps to also include situations where an answer is given that is inconsistent with some of the conceptual and foundational knowledge surrounding the question.

Performance Task Rubric

Here we provide examples for how Performance Tasks are graded. We chose problems from calculus, hoping that you will get the general idea without being confused about the mathematics.

Question.

Determine the area between the curve $y = \sin(x)$ and $y = \cos(x)$ over the x -range of $[0, \pi/4]$.

As a grader, I anticipate a student will convert the question into an integral calculation by subtracting one function from the other. Then, they will carry out the integral calculation by finding antiderivatives.

Solution #3. We find the area by calculating the integral

$$\int_0^{\pi/4} \sin(x) - \cos(x) dx = 1 - \sqrt{2} \approx -0.414\dots$$

Since areas are positive, the answer is 0.414...

Grading solution #3: 5/6.

- *Concept.* 1/2. The solution finds an integral but it has misidentified the difference to integrate and there is no indication the solve realizes the issue.
- *Mechanics.* 2/2. The integral is calculated just fine.
- *Grade-ability:* 1/1. The writing (typing) is neat.
- *Presentation:* 1/1. The writing narrates in a clear way how the question is being answered and what the outcome is.

You may want to compare with Solution #2. The solutions are similar. Here, the mechanics are perfect and there they are not. In conceptual terms, both solutions have issues. In this #3, you might argue there is more conceptual gaps. That's okay. My rubric rewards flawless aspects of the work while being generous with mistakes. Similar mistakes do not count double in my grading scheme. Read carefully how you earn 1/2 on the concept or mechanics points.

Remember: your professors have enormous agency in how they design grading. Grading is not objective, even if someone has told you otherwise. To mitigate that challenge, my rubric is designed to prioritize being transparent and upfront.

Solution #4. The area is given by

$$\sin(\pi/4) - \cos(\pi/4) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0.$$

Grading Solution #4: 3/6.

- *Concept.* 0/2. The solution does not point to a productive strategy to answer the question as given. (Plus, it is a conceptual issue to get area = 0.)
- *Mechanics.* 1/2. The calculation is literally correct but it overly simple.
- *Grade-ability:* 1/1. The writing (typing) is neat.
- *Presentation:* 1/1. The answer is clear in what it claims has happened.

Read. Check Revisions

Your Reading Checks will feature both *multiple* choice and *select* all questions. Depending on which type you revise, your revision will ask more or less of an explanation. Let's see how this look in practice.

Question. (Multiple choice) Which of the following indefinite integrals are calculated accurately?

(a) $\int \sin(x) dx = \cos(x) + C.$

(b) $\int \cos(x) dx = \sin(x) + C.$

(c) $\int \tan(x) dx = \sec^2(x) + C.$

Revision (1/1). The integral (b) is accurate because I looked it up on pg. 431. Since this is a multiple choice question, I do not need to address options (a) and (c).

Tip: You can justify your statement from a trusted source. (By the way, pg. 431 is just a random number.)

Question. (Select all that apply) Which of the following indefinite integrals are calculated accurately?

(a) $\int \cos(x) dx = \sin(x) + C.$

(b) $\int \sin(x) dx = \cos(x) + C.$

(c) $\int \tan(x) dx = \sec^2(x) + C.$

Revision (1/1). The integral (a) is accurate but (b) and (c) are not. The answer (a) is true because I looked it up on pg. 431 of our book. Next, (b) and (c) are false because it also says $\int \sin(x) dx = -\cos(x) + C$ and $\int \tan(x) dx = \ln(|\sec(x)|) + C.$

Tip. To show something is false, you can provide a corrected statement.

Question. (Select all that apply) Suppose $y = f(x)$ and $y = g(x)$ are two curves lying over x in $[a, b]$. Which of the following statements are accurate?

(a) The area between the two curves is always non-negative.

(b) The integral $\int_a^b f(x) - g(x) dx$ represents the area between the two curves.

(c) The integral $\int_a^b f(x) - g(x) dx$ is always non-negative.

(d) The integral $\int_a^b |f(x) - g(x)| dx$ represents the area between the two curves.

Revision (1/1). The answers (a) and (d) are accurate, while (b) and (c) are not. For (a), areas are always non-negative so that is easy. For (d), this integral is how you calculate the area (we learned this in the reading on pg. 109). Now we deal with (b) and (c). Suppose $f(x) = x$ and $g(x) = 2$ over the interval $[0, 1]$. Then

$$\int_0^1 x - 2 dx = \frac{1}{2} - 2 = -1.5.$$

So, (c) fails. The main issue is $g(x) = 2$ lies above $f(x) = x$ on $[0, 1]$ and so $\int_0^1 f(x) - g(x) dx$ calculates the *negative* of the area in this case. This also explains why (b) is false: the integral should have an absolute value symbol in it.

Tip: To show something is *false*, you can also provide a counter-example like we do here for cases (b) and (c).

Read. Check Revisions

Now let me illustrate how you may not get full points back on Reading Check revisions.

Question. (Multiple choice) Which of the following indefinite integrals are calculated accurately?

(a) $\int \sin(x) dx = \cos(x) + C.$

(b) $\int \cos(x) dx = \sin(x) + C.$

(c) $\int \tan(x) dx = \sec^2(x) + C.$

Revision (0/1). The integral (c) is accurate because I looked it up on pg. 431.

Logic: The revision is just wrong. You will know the answers before you revise, so I do not give credit for wrong answers, no matter the justification.

Question. (Select all that apply) Which of the following indefinite integrals are calculated accurately?

(a) $\int \cos(x) dx = \sin(x) + C.$

(b) $\int \sin(x) dx = \cos(x) + C.$

(c) $\int \tan(x) dx = \sec^2(x) + C.$

Revision (0.5/1). The integral (a) is accurate. The answer (a) is true because I looked it up on pg. 431 of our book.

Logic. The revision is partially right, but this is a select all question and so the revision process asks the student to also explain how they know (b) and (c) are not right answers.

Question. (Select all that apply) Suppose $y = f(x)$ and $y = g(x)$ are two curves lying over x in $[a, b]$. Which of the following statements are accurate?

(a) The area between the two curves is always non-negative.

(b) The integral $\int_a^b f(x) - g(x) dx$ represents the area between the two curves.

(c) The integral $\int_a^b f(x) - g(x) dx$ is always non-negative.

(d) The integral $\int_a^b |f(x) - g(x)| dx$ represents the area between the two curves.

Revision (0.5/1). The answers (a) and (d) are accurate, while (b) and (c) are not. For (a), areas are always non-negative so that is easy. For (d), this integral is how you calculate the area (we learned this in the reading on pg. 109). The reason (b) is wrong is because the absolute value in (d) needs to be there. The reason (c) is wrong is that the integral could be negative, if the area between the curves was negative.

Logic. The issue with this revision is that final phrase "...if the area between the curves was negative." That just is not an accurate statement, since area must be non-negative. I will read the revisions closely and parts of the explanation that are off like this will lead to less than full marks. (This is also why I recommend as on pg. 12 that you try to provide concrete examples to disprove statements.)

Perf. Task Revisions

Now let me illustrate revising a Performance Task might go. If you have any questions when revising your work, please come and find me or email me to ask your questions!

Question.

Determine the area between the curve $y = \sin(x)$ and $y = \cos(x)$ over the x -range of $[0, \pi/4]$.

Original Solution #1.

$$\cos(x) \geq \sin(x)$$

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = \sin(x) + \cos(x) \Big|_0^{\pi/4} = \sqrt{2} - 1 \approx 0.414...$$

Revision.

Original score: 5/6.

New solution. When x is in $[0, \pi/4]$ we have $\cos(x) \geq \sin(x)$. To calculate the area, we therefore subtract the larger one from the smaller and determine the integral over $[0, \pi/4]$. What we get is

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = \sin(x) + \cos(x) \Big|_0^{\pi/4} = \sqrt{2} - 1 \approx 0.414...$$

And therefore the area is 0.414.

Grader note: This revision would earn 6/6. And I am writing a few more words than you may, but the point is that the various steps have to be connected with some kind of words so that the grader can easily follow your work. Note also...please include your original score!

Original Solution #4. The area is given by

$$\sin(\pi/4) - \cos(\pi/4) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0.$$

Revision.

Original score: 3/6.

New solution. First, I should take the different in the right order. We have first

$$\cos(\pi/4) - \sin(\pi/4) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0.$$

I forgot to also calculate $\cos(0) - \sin(0) = 1$. Then I take the second answer and subtract the first. I get the area is 1.

Grader note: This revision would earn 4/6. The solver has included conceptual information that shows they understand there is something going on with calculating differences of functions. However, conceptual and mechanical gaps remain. In particular, there is no integral still, so the calculations are far too simple. Note: you may not earn back full points on Performance Tasks. Note also...please include your original score!

Generative AI Usage



Generative AI tools such as ChatGPT are powerful tools for learning. In my course, you are encouraged to responsibly use those tools to better your understanding.

Basic expectations

- **Reading Checks.** Complete Reading Checks *without* generative AI.
- **Exercise Sets.** Complete Exercise Sets *without* generative AI.

Details. Learning is messy. It has neither start nor finish. AI is just one tool you may use to better understand the material, and it's not like its going away. I've used it to help me study German, for instance. Students often use it to summarize or reexplain ideas from their lessons. If you do that in this course, let me know what you find! I designed this course knowing about generative AI capabilities. The revision system, in particular, is meant to relieve the pressure some you may feel to take learning shortcuts. The weight of each task in your final grade is also meant to deincentive systematic shortcut taking.



Your “at home” work consists of Reading Checks and Exercise Sets. In the next two columns I describe fair and unfair use of generative AI, in this course.

Fair use

- You **may** use generative AI before or after reading to reinforce your understanding.
- You **may** use generative AI to help explain a Reading Check question once it is graded.
- You **may** use generative AI to generate examples similar to Exercise Set problems.
- You **may** use generative AI to help you find mistakes in the graded work.
- You **may** *model* your language usage on the usage of of a generative AI assistant.

Unfair use

- You **may not** directly ask generative AI to answer your Reading Check questions.
- You **may not** use generative AI to generate revisions to Reading Check questions.
- You **may not** directly ask generative AI to answer Exercise Set problems for turn in.
- You **may not** use generative AI to generate revisions to Exercise Sets.
- You **may not** copy outputs from generative AI assistants (or anywhere else).

Temporary 0. Your written work ***must be in your own words and reflect your own understanding***. If a student turns in an answer that uses approaches, results, terminology, etc. that are not aligned with the context of the course, they will receive a “temporary 0” on the problem. The 0 can be removed upon meeting with me to discuss the origin and logic of the solution. Note: a written revision is not available for a solution I put a temporary 0 on. (Repeated from p. 5 for emphasis.)

Generative AI, cont.



The purpose of this page to provide a warning that using generative AI in my grading scheme may lead only to B-level grades, and that is only for the 35% of your grade based on Reading Checks and Exercise Sets. The learning gap AI creates with frequent use for shortcuts will not be worth it. Consider the question we used on p. 10-11.

Question.

Determine the area between $y = \sin(x)$ and $y = \cos(x)$ over x in $[0, \pi/4]$.

Answer from Chat GPT (prompt given Aug 3, 2025, edited for formatting).

We will do this in four steps.

1. Determine which function is on top.

- At $x = 0$, we have $\sin(0) = 0$ and $\cos(0) = 1$, so $\cos(x)$ is on top.
- At $x = \pi/4$ we have $\sin(\pi/4) = \cos(\pi/4) = \frac{\sqrt{2}}{2}$, so they are equal.

Therefore $\cos(x)$ is on top of $\sin(x)$.

2. Set up the area integral. We have $(\text{Area}) = \int_0^{\pi/4} \cos(x) - \sin(x) dx$.

3. Solve the integral. We have

$$\int_0^{\pi/4} \cos(x) - \sin(x) dx = \sin(x) + \cos(x) \Big|_0^{\pi/4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - (0 + 1) = \sqrt{2} - 1.$$

4. The answer is $(\text{Area}) = \sqrt{2} - 1$.

Grading this solution: 5/6.

- *Concept.* 2/2. The solver orders the two functions, convert the area into an integral and then solve the integral.
- *Mechanics.* 1/2. There is a mechanical gap in Step #1. The solver only checks that $\cos(x) \geq \sin(x)$ for the two endpoints $x = 0$ and $x = \pi/4$. The mechanics says you need to justify it for all values of x , otherwise you risk that $y = \cos(x)$ crosses $y = \sin(x)$ somewhere in the middle of the range.
- *Grade-ability:* 1/1. The writing (typing) is neat.
- *Presentation:* 1/1. The writing narrates in a clear way how the question is being answered and what the outcome is.

Note. The example question was on my syllabi last year also, but the AI pages were added this year. I did not “rig” the example to create a gap. ChatGPT did that all on its own! I have seen gaps like this a lot when asking ChatGPT questions starting at the 2xxxx-level in the University curriculum. The primary flaw in overusing generative AI is mistaking its confidence for correctness. Generative AI tools lack true intelligence and logical or learning capabilities. They are language models that guess answers based on statistics and probability. They do not use deductive reasoning and they do not reference or check background knowledge. They are often wrong about basic work and will give you absolutely no indication there is a gap.